Smart DNS Traffic Analyzer

REAL-TIME THREAT DETECTION VIA CLI & Tkinter

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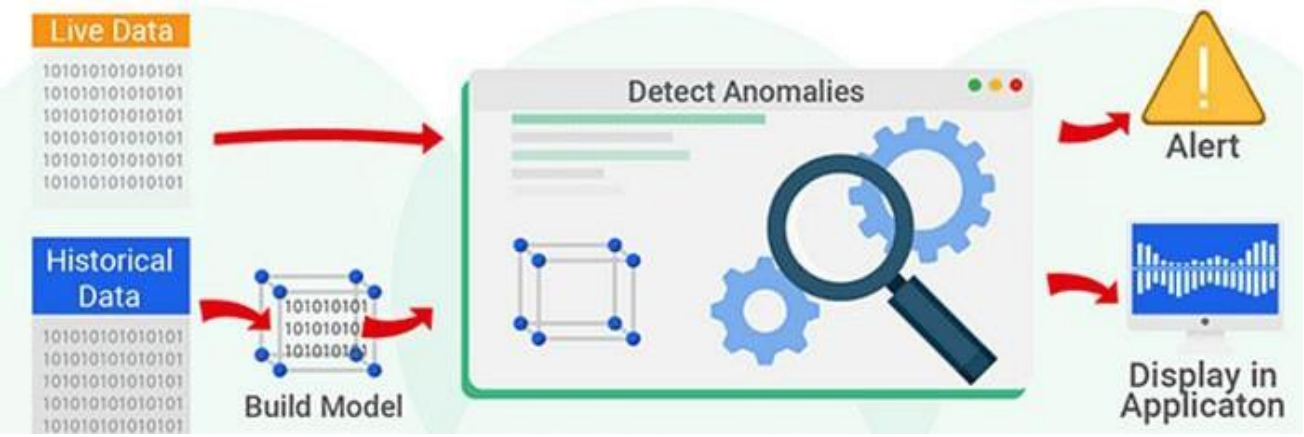
Project Category

Cybersecurity – Network security and anomaly detection



This project introduces a command-line tool designed to monitor DNS traffic in real-time and detect threats like tunneling and spoofing. Built for interns like us, it's a simple yet powerful way to learn and protect networks without needing fancy interfaces.

Problem Statement

DNS, the system that turns website names (like google.com) into computer addresses, is often a target for hackers. They use tricks like **DNS tunneling** to sneak data out or **spoofing** to send you to fake sites. These attacks can steal information, spread malware, or trick users into giving away passwords.

For exampe, a hacker might hide stolen files in a long, weird domain name or fake a bank's DNS reply to look legit. Current tools to catch these are often complex, expensive, or need special screens, making them hard for interns to use. We need a fast, free tool that works right in the terminal, where we can see threats as they happen and learn how to stop them.

This problem affects small companies and learners alike. Without real-time detection, attacks go unnoticed until it's too late. Our tool fills this gap with a command-line approach, perfect for hands-on practice.

Objectives

The **Smart DNS Traffic Analyzer** is designed to enhance **DNS security monitoring** by leveraging **network analysis and machine learning** to detect cyber threats. The key objectives of this project are:

1 Real-Time DNS Monitoring

- Capture and analyze **live DNS traffic** to track network activity.
- Identify patterns in DNS queries to detect suspicious behavior.

2 Threat Detection & Prevention

- Detect **DNS tunneling, spoofing, and C2 communication** used by attackers.
- Use **machine learning** to classify DNS queries as **safe or malicious**.
- Integrate with **threat intelligence feeds** (e.g., VirusTotal) to validate threats.

3 User-Friendly Command-Line Tool

- Ensure easy deployment and usage with a **lightweight CLI-based tool**.
- Provide **clear alerts and logs** for security teams to take action.

4 Skill Development & Learning

- Help **security analysts and interns** understand **DNS-based attacks**.
- Allow users to experiment with **real-world network threats** in a controlled environment.

5 Foundation for Advanced Security Solutions

- Act as a **building block** for larger **SIEM, IDS, and network monitoring** systems.
- Encourage further research in **AI-based anomaly detection in DNS traffic**.

This project **empowers cybersecurity professionals** to proactively detect DNS-based threats, **strengthens network security**, and provides a **practical learning tool** for security enthusiasts.

Scope of Work

The **Smart DNS Traffic Analyzer** is a command-line tool designed to **monitor DNS traffic** and detect **tunneling and spoofing attacks in real-time**. It is lightweight, cross-platform, and built for **security analysts and interns** to gain hands-on experience in **DNS-based threat detection**.

🔹 **What This Tool Does:**

✅ **Captures live DNS queries and responses** to analyze network behavior.
✅ **Flags suspicious activity** (e.g., high-entropy domains, frequent lookups).
✅ **Logs findings in real-time** with detailed alerts for further investigation.
✅ **Lets users tweak detection thresholds** for entropy, query frequency, and response anomalies.

🔹 **What This Tool Doesn't Do (Yet):**

❌ **No Threat Intelligence API Integration** (e.g., VirusTotal, OpenDNS).
❌ **Limited Scalability** – May struggle with very high network traffic.
❌ **Partial DNS Threat Coverage** – Detects tunneling and spoofing but not full phishing protection or DNS amplification attacks.
❌ **Basic Spoofing Detection** – Doesn't yet verify DNS poisoning attacks comprehensively.

🔹 **Limitations & Future Scope**

🔹 **Small Training Dataset** – Enhancing ML-based anomaly detection will improve accuracy.
🔹 **Advanced Attack Detection** – Expanding to detect **fast-flux botnets and exfiltration techniques**.
🔹 **SIEM & Threat Feed Integration** – Future versions may connect to **Splunk, ELK, or Open Threat Exchange**.

Scope Diagram:

[Monitor] → **[Detect]** → **[Log]**

(Live Capture) (Threat Analysis) (Real-Time Alerts)

Methodology & Approach

The **Smart DNS Traffic Analyzer** is a real-time command-line tool designed to **monitor, analyze, and detect** suspicious DNS activity using **machine learning and rule-based detection**. The tool evaluates **entropy, length, TTL, frequency, and subdomains** to classify DNS queries as **normal or malicious**.

1. Capturing DNS Traffic

- The tool **sniffs live DNS packets** using **Scapy**.
- It listens on **port 53 (UDP & TCP)** to track all DNS queries and responses.
- Captured queries are analyzed in real-time to detect tunneling and spoofing attempts.

Outcome: Continuous monitoring of DNS traffic for security analysis.

2. Feature Extraction & Threat Detection

- The tool extracts key DNS features: **query length, entropy, subdomains, TTL, and frequency**.
- High entropy and long domain names may indicate **DNS tunneling**.
- Low TTL values may signal **DNS spoofing**.
- Frequent lookups in short intervals could indicate **C2 communication**.

Outcome: Suspicious queries are flagged based on predefined security rules.

3. Machine Learning & Rule-Based Detection

- Uses a **Random Forest model** trained on labeled DNS data.
- If a domain exceeds entropy or length thresholds, it is immediately flagged.
- Spoofing detection is done using TTL values.

Outcome: The tool predicts **DNS tunneling attacks** and detects spoofing attempts.

4. Logging & Alerts

- **Terminal Output:** Displays live alerts with **color-coded risk levels**.
- **dns_report File (**dns_report_0000.txt**)**: Saves all flagged queries for later analysis.
- **Allowlist/Blocklist Integration:** Users can define trusted and blacklisted domains.

Outcome: Security teams can analyze DNS logs and adjust detection thresholds.

How to Run It:

- Run the requirement.txt file before the running the code

  pip install -r requirement.txt

- Windows: Open Command Prompt as Administrator:

  ```
  cd C:\SmartDNSTrafficAnalyzer
  python smart_dns_analyzer.py
  ```

  Or with custom thresholds:

  ```
  python smart_dns_analyzer.py --entropy 3.5 --length 30 --ttl 10 --freq 5
  ```

- Kali Linux: Open Terminal with sudo:

  ```
  cd /path/to/SmartDNSTrafficAnalyzer
  sudo python3 smart_dns_analyzer.py
  ```

  Or with custom thresholds:

  ```
  sudo python3 smart_dns_analyzer.py --entropy 3.5 --length 30 --ttl 10 --freq 5
  ```

- macOS: Open Terminal (admin via sudo if needed):

  ```
  cd /path/to/SmartDNSTrafficAnalyzer
  python3 smart_dns_analyzer.py
  ```

Create Allowlist/Blocklist (Optional)

- In the same folder, create `allowlist.txt` (e.g., `www.google.com`) and `blocklist.txt` (e.g., `badsite.com`).
- Use **Notepad (Windows)** or **nano (Linux/macOS)** to edit these lists.

Outcome: The tool will allow/block specific domains based on user-defined lists.

Key Parameters

The tool evaluates multiple DNS characteristics to detect malicious behavior. The following parameters help classify DNS traffic:

Parameter	Description	Normal	Suspicious
Length	Counts characters in a domain. Longer domains may indicate tunneling.	≤ 30	> 30
Entropy	Measures randomness; high entropy suggests encoded or malicious data.	≤ 3.5	> 3.5
Subdomains	Counts dots in the domain. Too many suggest abnormal behavior.	≤ 3	> 3
TTL	Checks response time; low TTL may indicate spoofing or fast-flux.	≥ 10	< 10
Frequency	Counts query repetitions in 10 seconds. High frequency may indicate C2.	≤ 5	> 5
Status	Overall classification of the query.	Normal	Suspicious/ Spoofing

Adjustable Settings:

- Users can customize detection thresholds via **command-line arguments** to match their network environment.

Tools & Technologies

The **Smart DNS Traffic Analyzer** is built using **lightweight, open-source technologies** to ensure ease of use, flexibility, and powerful DNS monitoring capabilities.

🔹 Core Technologies Used

Tool	Purpose
Python	The main programming language, simple and beginner-friendly.
Scapy	Captures and analyzes DNS packets, acting like a network spy.
Pandas	Organizes, structures, and processes DNS data for analysis.
Scikit-learn	Provides machine learning algorithms for threat detection.
Logging	Stores DNS queries and analysis results in log files.
tkinter	Allows users to create graphical user interfaces for desktop applications.
Npcap	Used to capture the DNS network packet for analysis and monitoring.

🔹 Why These Tools?

- **Beginner-Friendly** – Python makes it easy for interns to modify and learn.
- **Efficient Network Monitoring** – Scapy provides real-time DNS capture.
- **Machine Learning Ready** – Scikit-learn enables AI-based threat detection.
- **Customizable & Free** – Users can tweak detection rules without extra cost.

🔹 System Requirements

OS Compatibility: Windows, Linux (Kali), macOS
Python Version: 3.x
Dependencies: Install via:

```
pip install scapy pandas scikit-learn beepy
```

Permissions: Run with **Admin/Sudo** for network monitoring.

Roles & Responsibilities

This project was a collaborative effort where each team member played a crucial role in successfully developing and documenting the **Smart DNS Traffic Analyzer**.

Role	Team Member	Responsibilities
Project Leader	*Manish Pandey*	Researched DNS security, planned project workflow, and guided the team.
Security Analyst	*Harshil Shiroya*	Analyzed threats, interpreted DNS logs, and prepared documentation.
Developer 1	*Pallavi Kadam*	Implemented core code based on given instructions.
Developer 2	*Manish Pandey*	Assisted in coding, debugging, and improving detection features.
Presentation Lead	*Vaishnavi chavan*	Designed the PowerPoint presentation and will present the project to the class.

Expected Outcomes

The **Smart DNS Traffic Analyzer** is designed to enhance **network security** and provide a **practical learning experience**. The key expected outcomes of this project include:

1 Real-Time Threat Detection

- The tool will **detect DNS tunneling and spoofing attacks** as they happen.
- It will help **secure small networks** by flagging suspicious DNS activity.
- Users can **identify malicious queries** and take immediate action.

2 Hands-On Learning

- Develops **practical skills** in detecting DNS-based cyber threats.
- Enables **real-time monitoring** to analyze live DNS traffic.
- Allows users to **simulate and test network attacks** in a controlled environment.

3 Log-Based Threat Analysis

- The tool **stores logs** for further **review and pattern analysis**.
- Security teams can **examine past threats** and improve defense strategies.
- Logs help **detect recurring malicious domains** for proactive security.

4 Real-World Application Scenario

- **Example:** An intern runs the tool on a **company network** and detects **DNS tunneling** from an unknown domain.
- The logs confirm repeated lookups to **a suspicious server**, indicating data exfiltration.
- The intern **blocks the domain**, preventing a potential **cyberattack** before damage occurs.

Outcome: This tool **not only detects threats but also empowers users** to act against them in **real-world scenarios**.

Deliverables & Documentation

The **Smart DNS Traffic Analyzer** project includes the following key deliverables:

1 Tool Script

File: smart_dns_analyzer.py
✅ This is the **main script** that captures DNS traffic, analyzes queries, and detects tunneling or spoofing attacks.
✅ Users can **run it directly** to start real-time monitoring and adjust thresholds as needed.

2□ Report File

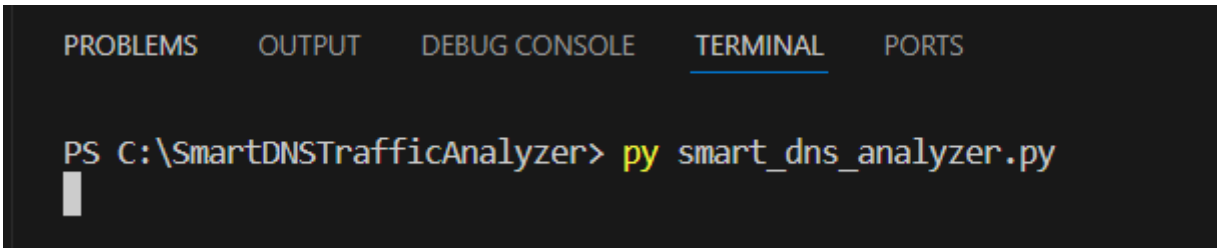
File: dns_report_0000.txt
✅ Stores **detailed analysis of all captured DNS queries**.
✅ Helps track **suspicious patterns, flagged domains, and network threats** over time.
✅ Can be used for **further investigation** and security audits.

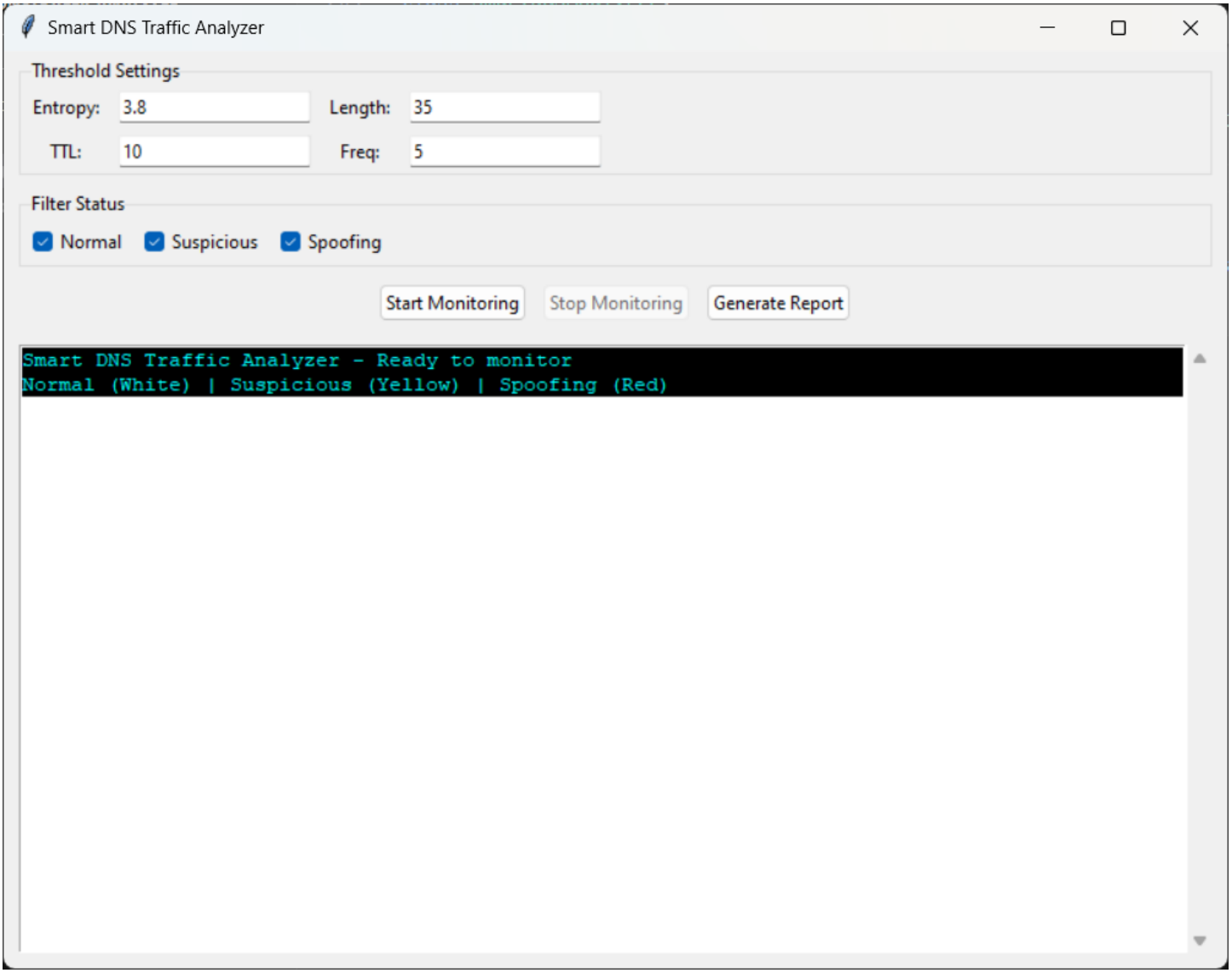
3□ Project Report

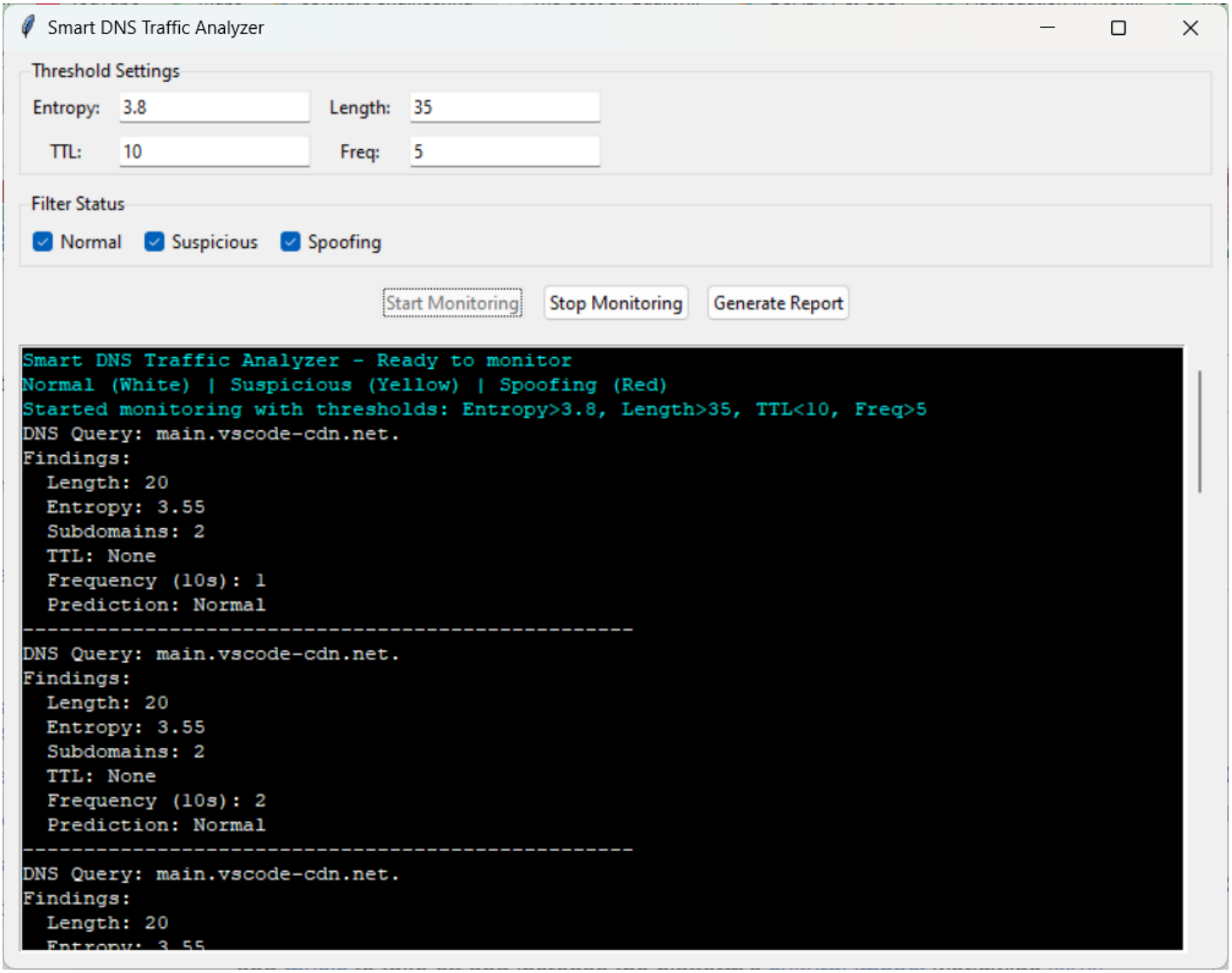
Document: This report (22+ pages)
✅ Provides **step-by-step guidance** on setting up, running, and understanding the tool.
✅ Covers **methodology, features, expected outcomes, and real-world applications**.

4 **Review the logs:** Open dns_report_0000.txt to analyze flagged queries and threats.

Screenshot :


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\SmartDNSTrafficAnalyzer> py smart_dns_analyzer.py


Smart DNS Traffic Analyzer
Threshold Settings
Entropy: 3.8
Length: 35
TTL: 10
Freq: 5
Filter Status
Normal
Suspicious
Spoofing
Start Monitoring
Stop Monitoring
Generate Report
Smart DNS Traffic Analyzer - Ready to monitor
Normal (White) | Suspicious (Yellow) | Spoofing (Red)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Generate Report
Smart DNS Traffic Analyzer - Ready to monitor
Normal (White) | Suspicious (Yellow) | Spoofing (Red)
Started monitoring with thresholds: Entropy>3.8, Length>35, TTL<10, Freq>5
DNS Query: main.vscode-cdn.net.
Findings:
Length: 20
Entropy: 3.55
Subdomains: 2
TTL: None
Frequency (10s): 1
Prediction: Normal

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